

### Cell Biology & Biochemistry

**Topics Covered**

**Biochemistry**

- Biomolecules and Molecular Organization
- Biological Macromolecules:
  - Carbohydrates
  - Lipids
  - Proteins
  - Nucleic acids and Nucleotides
  - Identification of Biomolecules (Practical)

**Cell Biology**

- Cell ultra-structure
- Biological membranes: Structure and functions
- DNA Replication
- Protein Biosynthesis
- Cell Division
- Cell signaling

**Bioenergetics**

- Enzymes: General properties and mechanism of action
- Cellular respiration

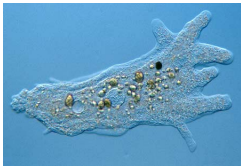
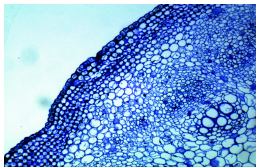


**BIO 1201**  
**Cell Biology & Biochemistry**  
**Lesson 5 – Cell Ultrastructure**

### Cell Biology

**Cells**

- All organisms are composed of cells
- Cells - basic structural, functional, and biological unit of all living organisms
  - smallest unit of life that can replicate independently
- Some organisms consist of a single cell, others many
- Cell biology – study of internal structure and function of cells

### Cell Biology

**A brief history of Cell Biology**

- Robert Hooke – observed dead plant cells in 1665
- Antonie van Leeuwenhoek – observed first live cells (animalcules) in 1670s
- Matthias Schleiden (1838) – all plants consists of cells
- Theodor Schwann (1839) – all animal tissues consists of cells
- Rudolf Virchow (1855) – All cells arise from preexisting cells



### Cell Biology

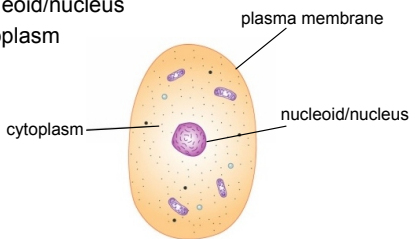
**Cell theory**

1. All organisms are composed of one or more cells and life processes of **metabolism** and **heredity** occur within the cells.
2. Cells are the smallest living things, the basic units of organization of all organisms.
3. Cells arise only by division of a previously existing cell.

### Cell Biology

**Cells**

- Three major features of cells
  - plasma membrane
  - nucleoid/nucleus
  - cytoplasm

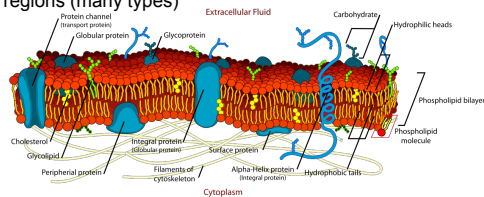


## Cell Biology

### Cells

#### Plasma membrane

- Encloses the cell, separates contents from its' surrounding
- Consists of a glycerophospholipid bilayer
- 5-10 nm thick
- Proteins are embedded in the glycerophospholipid bilayer
- Proteins also have large hydrophobic and hydrophilic regions (many types)

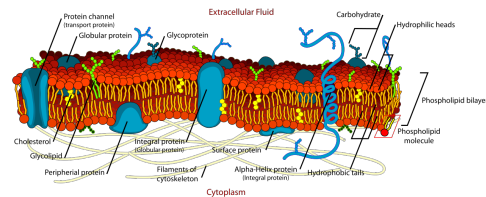


## Cell Biology

### Cells

#### Plasma membrane

- Transport Proteins – transport molecules across the membrane
- Receptor Proteins - induce changes within the cell when they come in contact with specific molecules in the environment

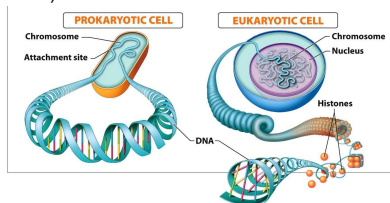


## Cell Biology

### Cells

#### Nucleus/Nucleoid

- Every cell contains hereditary material – DNA
- Prokaryotes – single circular DNA molecule (**Nucleoid**)
- Eukaryotes – DNA is surrounded by two membranes (**Nucleus**)

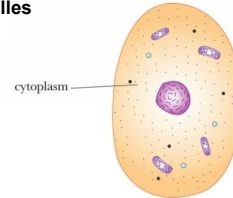


## Cell Biology

### Cells

#### Cytoplasm

- **Semifluid** matrix that fills the interior of the cell
- Contains – sugars, nucleotides, amino acids, proteins, minerals (Cytosol)
- May have specialized membrane bound compartments - **organelles**



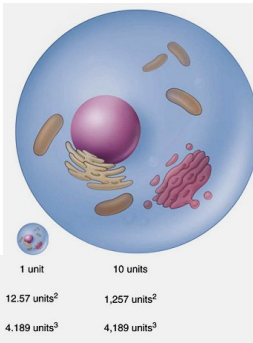
## Cell Biology

### Cells

#### Cell size

- Generally 50  $\mu\text{m}$  in diameter
- Substances enter and leave the cell through simple diffusion
- Smaller cells have greater surface area to volume ratio
- Larger cells take more time for diffusion

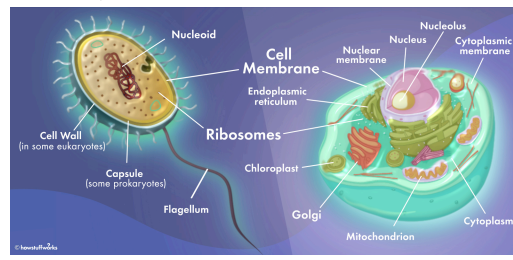
| Cell radius (r)                 | 1 unit                   | 10 units                 |
|---------------------------------|--------------------------|--------------------------|
| Surface area ( $4\pi r^2$ )     | 12.57 units <sup>2</sup> | 1,257 units <sup>2</sup> |
| Volume ( $\frac{4}{3}\pi r^3$ ) | 4.189 units <sup>3</sup> | 4,189 units <sup>3</sup> |



## Cell Biology

### Cells

- Two major types
  - Prokaryotic
  - Eukaryotic



## Cell Biology

### Cells

#### Ribosomes

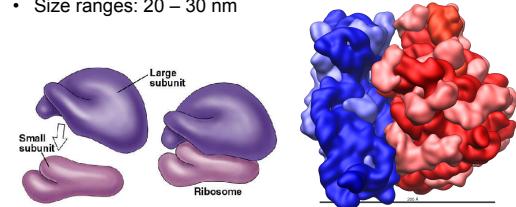
- Organelles that are made of RNA (ribosomal RNA) bound within a complex of several different proteins
- Involved in protein synthesis
- Consists of two subunits (small, large)
- Subunits join to form a functional ribosome only during protein synthesis
- Metabolically active eukaryotic cells may have several million ribosomes in each
- Size of ribosomes - Svedberg units (S)
- Prokaryotes – 70S, Eukaryotes – 80S

## Cell Biology

### Cells

#### Ribosomes

- Size ranges: 20 – 30 nm



## Cell Biology

### Cells

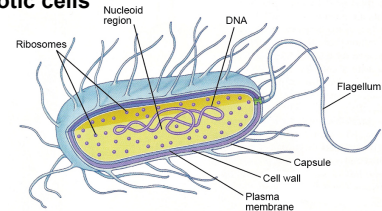
#### Ribosomes

- Prokaryotic ribosomes 20 nm, comprise 65% rRNA, 35% proteins
- Eukaryotic ribosomes 25-30 nm, comprise equal proportions of RNA and proteins
- Proteins act as a scaffold on ribosomes



## Cell Biology

### Prokaryotic cells



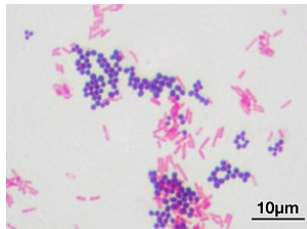
- Small cells (1-10  $\mu\text{m}$ ) surrounded by a membrane
- Encased with a rigid cell wall
- No interior compartments
- Found in single cells or chains
- Cell wall – Peptidoglycan (polymer of sugars and amino acids)

## Cell Biology

### Prokaryotic cells

- Two types of prokaryotic cells based on the Gram stain
  - Gram positive (stained by the Gram stain)
  - Gram negative (not stained by the Gram stain)

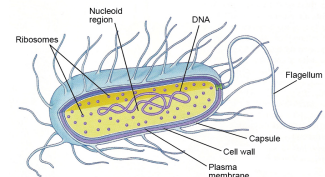
- Some prokaryote cells (e.g. bacteria) are covered by polysaccharides (capsule)



## Cell Biology

### Prokaryotic cells

- No membrane bound organelles
- Ribosomes present in cytoplasm
- A nucleoid present (simple circular DNA)
- Photosynthesis – takes place in folded plasma membrane that bears photosynthetic pigments
- Locomotion – flagellum/flagella
- Flagellum – thread like structure, extension of the cell membrane, made of protein fibers that are extended from the cell



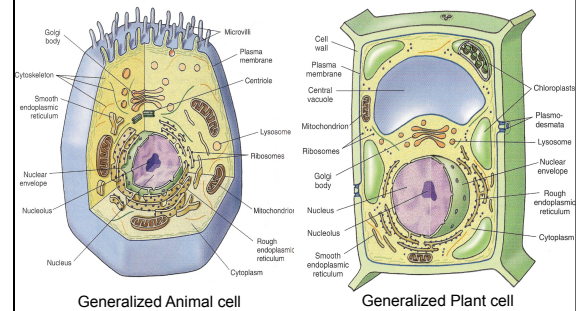
## Cell Biology

### Eukaryotic cells

- Larger than prokaryotic cells (10-100  $\mu\text{m}$ )
- Shows **compartmentalization** through organelles
- Organelles - membrane bound structures that allow multiple biochemical processes to proceed at the same time independently
- **Nucleus** - organelle containing DNA wrapped tightly around proteins (packed into chromosomes)
- Cytoskeleton – contains protein fibers that provide support
- Vacuoles – only in plant cells, stores proteins, pigments, waste materials, regulate hydrostatic pressure
- Cell wall – only in plants, fungi and some protist cells

## Cell Biology

### Eukaryotic cells

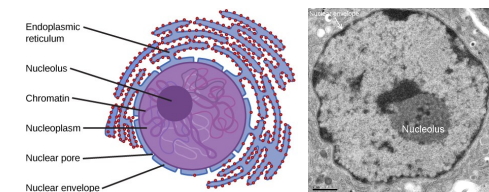


## Cell Biology

### Eukaryotic cells

#### Nucleus

- Largest organelle in the cell, generally a single nucleus
- Shape – roughly spherical (globular)
- Cytoskeleton holds the nucleus in place

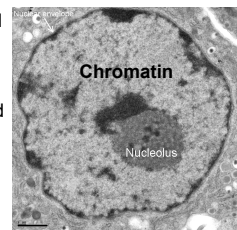


## Cell Biology

### Eukaryotic cells

#### Nucleus

- DNA – carries the hereditary information specifying cell structure & function
- DNA in the nucleus is organized into chromosomes
- Normally (when not dividing) they are fully extended to thread like strands – **chromatin**
- Nucleolus - site of ribosome synthesis in the nucleus



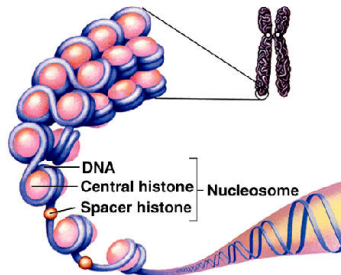
Nucleus (TEM)

## Cell Biology

### Eukaryotic cells

#### Nucleus

#### Organization of DNA in the chromosome



## Cell Biology

### Eukaryotic cells

#### Nucleus

- Nuclear envelope – two layers (each is a phospholipid bilayer)
- Nuclear pores – tiny openings in the nuclear envelope, regulates communication with the cytoplasm
- Pores are lined by protein molecules (nucleoporins)
- Pores allow the entry and exit of,
  - Proteins useful for nuclear structure/catalytic activity
  - RNA and ribosomes formed in the nucleus
  - small molecules and ions



Nucleus (SEM)



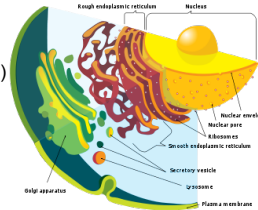
## Cell Biology

### Eukaryotic cells

Endomembrane system

**Set of membranes that form a single functional and developmental unit that are either connected directly or exchange material through vesicle transport**

- Include,
  - Nuclear membrane
  - Endoplasmic reticulum (ER)
  - Golgi apparatus
  - Vesicles
  - Endosomes
  - Cell membrane

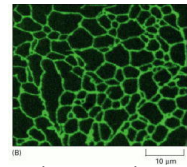


## Cell Biology

### Eukaryotic cells

Endomembrane system – Endoplasmic reticulum

- Composed of a lipid bilayer embedded with proteins
- Network of internal membranes – forms the compartments and vesicles
- Two types: RER and SER



Rough Endoplasmic reticulum (RER)

- The surface is studded with ribosomes – gives a rough appearance
- Proteins are synthesized on the surface

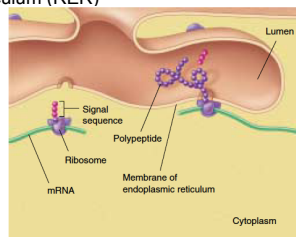
## Cell Biology

### Eukaryotic cells

Endomembrane system – Endoplasmic reticulum

Rough Endoplasmic reticulum (RER)

- Proteins are passed through the RER membrane and moved to a vesicle forming system – golgi apparatus



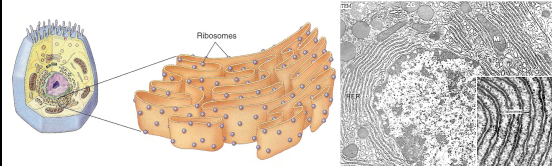
## Cell Biology

### Eukaryotic cells

Endomembrane system – Endoplasmic reticulum

Rough Endoplasmic reticulum (RER)

- Vesicles take the proteins
  - to the surface of the plasma membrane to be released to the outside of the cell
  - Golgi apparatus for further modification



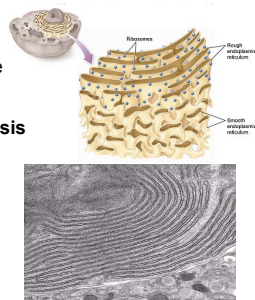
## Cell Biology

### Eukaryotic cells

Endomembrane system – Endoplasmic reticulum

Smooth Endoplasmic reticulum (SER)

- Few ribosomes on the surface
- Membranes have many embedded enzymes
- Enzymes catalyze the **synthesis of carbohydrates, lipids, phospholipids & steroids**
- Stores  $\text{Ca}^{+2}$
- Detoxify drugs (in the liver)
- Abundant in lipid metabolizing cells

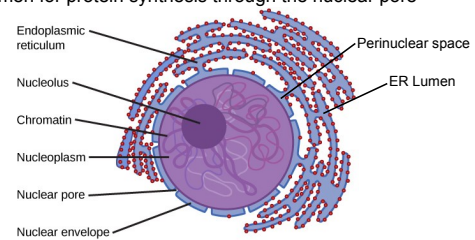


## Cell Biology

### Eukaryotic cells

Endomembrane system - Nuclear membrane

- Outer nuclear membrane is continuous with the ER lumen
- Synthesized mRNA in the nucleus can be sent to the ER lumen for protein synthesis through the nuclear pore

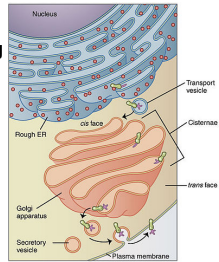


## Cell Biology

### Eukaryotic cells

#### Endomembrane system – Golgi apparatus

- Flattened stack of membranes
- Abundant in glandular cells
- Involved in **collection, packaging & distribution** of molecules synthesized in one place of the cell and utilized at another location
- Proteins & lipids synthesized in ER are **modified** in Golgi apparatus
- Proteins undergo **post-translational modification** e.g. Glycoproteins

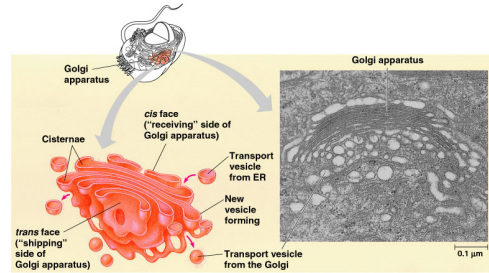


## Cell Biology

### Eukaryotic cells

#### Endomembrane system – Golgi apparatus

- Involved in the synthesis of plant cell wall

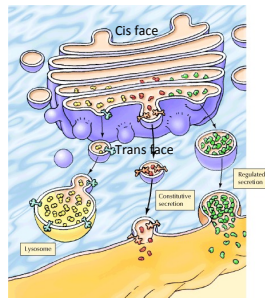


## Cell Biology

### Eukaryotic cells

#### Endomembrane system – Golgi apparatus

- Protein sorting and exportation
- Secretion
  - constitutive secretion
  - regulatory secretion



## Cell Biology

### Eukaryotic cells

#### Endomembrane system – Vesicles

- Small (0.1-1 μm) organelles within a cell, consisting of fluid enclosed by a lipid bilayer membrane
- Three major roles
  - transport molecules between organelles (transport vesicles)
  - storage of molecules for later use or secretion (secretory vesicles, vacuoles)
  - Digestion of cellular components and disposal of waste (Lysosomes, microbodies)

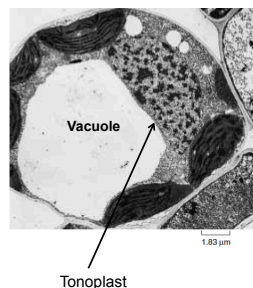
## Cell Biology

### Eukaryotic cells

#### Endomembrane system – Vesicles

##### Vacuoles

- Only found in plant cells, bacteria, fungi and protists
- Placed centrally in the cell
- Stores large amounts of water and sugars, ions, pigments and degradative enzymes
- Other functions
  - increase the surface Area-to-volume ratio of the plant cell by applying pressure to the cell membrane
  - degrade macromolecules
  - store water in plant cells
  - store pigments (anthocyanin)



## Cell Biology

### Eukaryotic cells

#### Endomembrane system – Vesicles

##### Lysosomes

- Membrane bound digestive vesicles, released by Golgi app.
- Contains digestive enzymes (Acid hydrolases) of the cell
- low pH (5), active enzymes
- high pH (>6), inactive enzymes
- Enzymes catalyze the rapid breakdown of proteins, nucleic acids, lipids, carbohydrates and old organelles

## Cell Biology

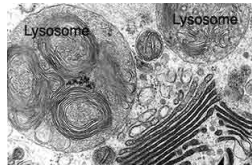
### Eukaryotic cells

Endomembrane system – Vesicles

Lysosomes

- Lysosomes maintain the higher interior pH by actively pumping  $H^+$  ions by a proton pump present on the membrane

- Lysosome degrade material from endocytosis, phagocytosis, autophagy

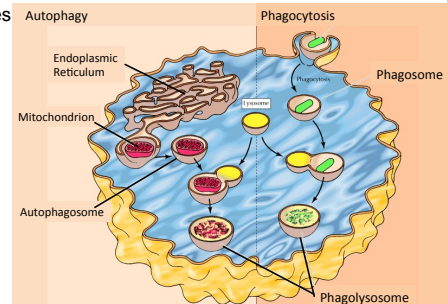


## Cell Biology

### Eukaryotic cells

Endomembrane system – Vesicles

Lysosomes



## Cell Biology

### Eukaryotic cells

Endomembrane system – Vesicles

Microbodies

- Enzyme bearing membrane bound organelles
- Grow by incorporating lipids & proteins and dividing
- Two types - Glyoxisomes
  - peroxisomes
- Glyoxisomes are found only in plant cells, convert fats into carbohydrates

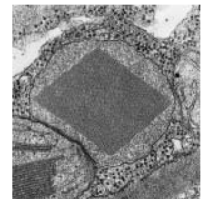
## Cell Biology

### Eukaryotic cells

Endomembrane system – Vesicles

Microbodies: Peroxisomes

- Found in all eukaryotic cells
- Spherical organelle
- Contains a large diamond shape protein crystal
- Has enzymes that catalyze the removal of electrons & associated hydrogen atoms (especially  $H_2O_2$ ) e.g. oxidase, catalase, peroxidase
- digestion of long chain fatty acids

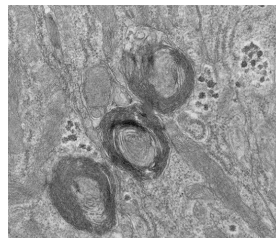


## Cell Biology

### Eukaryotic cells

New cell organelle discovered from fruit fly cells!

- PXo bodies
- Stores phosphate
- Purpose of storing phosphate is not known



Source: Xu et al. (2023) Nature. DOI: 10.1038/s41586-023-06039-y

## Cell Biology

### Summary

- Cells are the basic structural, functional, and biological unit of all living organisms
- Three major features of cells are the presence of a plasma membrane, nucleoid/nucleus and a cytoplasm
- Ribosomes are small structures found in all types of cells involved in protein synthesis
- Nucleus is the largest organelle in the cell and carries DNA which is the hereditary material of the cell
- Endomembrane system includes the nuclear membrane, endoplasmic reticulum, Golgi apparatus, vesicles, endosomes, cell membrane
- RERs are sites of protein synthesis destined for distribution
- SERs synthesize lipids, phospholipids and steroids

## Cell Biology

### Summary

- Golgi apparatuses are involved in collection, packaging and distribution of molecules
- Vesicles are membrane bound structures involved in storage, transportation and digestion of molecules inside the cell